

Weather Prediction Using Machine Learning

Muhammad Waleed Dar | AI/ML Intern | Phase 1 Report

1. Problem Statement

Accurate weather forecasting is critical for agriculture, disaster management, and transportation planning. Traditional forecasting relies on complex physical models that require extensive computational resources. This project explores whether Machine Learning classification models, trained on historical weather observations, can accurately predict whether it will rain the following day (**RainTomorrow**: Yes/No) — making forecasting more accessible and data-driven. This work builds on prior experience developing a Smart Weather Monitoring Station using Arduino and C++, extending that interest from live sensor-based measurement into predictive machine learning.

2. Dataset Overview

Property	Details
Source	Kaggle — Rain in Australia Dataset
Link	kaggle.com/datasets/jsphyg/weather-dataset-rattle-package
File	weatherAUS.csv
Rows	145,460
Columns	23 (16 numerical, 7 categorical)
Target Variable	RainTomorrow (Binary: Yes / No)
Date Range	2007 – 2017 (10 years, multiple AU locations)
Missing Values	Present in 21 / 23 columns
Class Distribution	No Rain: ~84%, Rain: ~16% (after preprocessing (Steps 1-10). After MinMax Normalization , prior to Feature/Target split and SMOTE-Tomek).

Dataset justification: weatherAUS.csv contains real-world observations with missing values, categorical features, and class imbalance — making it well-suited for a complete, realistic preprocessing pipeline rather than a pre-cleaned toy dataset.

3. Models to be Used (Upcoming Phases)

Model	Role
Logistic Regression	Baseline classifier
Decision Tree Classifier	Captures non-linear decision patterns
Random Forest Classifier	Primary model — ensemble of trees, reduces overfitting

Planned evaluation metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC.

4. Preprocessing Steps Performed (13 Steps)

Step	Description
Step 1	Dataset loading and shape verification (145,460 rows x 23 cols)
Step 2	Basic exploration: info(), describe(), head(), tail(), nunique()
Step 3	Duplicate check — 0 duplicates found
Step 4	Null value detection — missing in 21/23 columns
Step 5	Null handling — mode for categorical, median for numerical columns
Step 6	Label Encoding — 7 categorical columns converted to numeric
Step 7	Correlation matrix — feature relationships visualized via heatmap
Step 8	Dropped Date column — not relevant for ML training (not directly usable in raw form)
Step 9	Outlier detection & removal — IQR method on Rainfall column
Step 10	MinMax Normalization — numeric features scaled to [0,1] (target excluded)
Step 11	Class imbalance check — No Rain: 95,420 vs Rain: 18,228
Step 12	Feature/Target split — X (21 features), y (RainTomorrow)
Step 13	SMOTE-Tomek applied — balanced to 95,257 samples per class

Why outlier removal before scaling? MinMaxScaler sets its [0,1] range from each column's min/max; removing extreme Rainfall outliers (e.g. 371mm) first keeps those bounds reflective of everyday values instead of being stretched by rare storms. **Why SMOTE-Tomek?** The dataset was imbalanced (~84% No Rain vs ~16% Rain); a model trained on this directly would favor the majority class. SMOTE-Tomek oversamples the minority class with synthetic, interpolated samples while removing ambiguous borderline samples — producing a balanced dataset (95,257 samples per class).

5. Libraries & Tools

Language	Data Manipulation	ML & Preprocessing	Class Imbalance
Python 3	Pandas, NumPy	Scikit-learn	Imbalanced-learn (SMOTE-Tomek)
Visualization	Environment	Version Control	
Matplotlib, Seaborn	Jupyter Notebook, VS Code	Git & GitHub	

6. Challenges Encountered

High missing values: Sunshine (48%) and Cloud columns (38–41%) handled via median/mode imputation to preserve dataset size. **Severe class imbalance:** ~84% No Rain vs ~16% Rain — resolved via SMOTE-Tomek (synthetic oversampling + borderline cleaning). **Mixed data types:** 7 categorical columns required Label Encoding before numeric operations or model training.

7. GitHub Repository

github.com/muhammad-waleed-dar/Weather-Prediction-ML — notebook, dataset & report under **Phase1-EDA/**.